Quiz 7

Friday, October 23 — systems, elimination, and RREF (Week 8)

Last updated: October 8, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers the week before only: Systems, Elimination, RREF (Week 8). Same-day Lab 4 / dynamics are **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) · Lachniet Octave PDF · also on Resources.

Schedule

Core

Fluency (complete prompts — practice these by hand)

C7.1. Write as $Ax = b$ and classify square / over / under:

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ -x_1 + x_3 &= 1. \end{aligned}$$

C7.2. One elimination step: clear the $(2,1)$ entry of

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 1 & 0 \end{array}\right]$$

by row replacement. Write the new matrix.

C7.3. From the echelon form

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -3 & -6 \end{array}\right],$$

solve by back substitution.

C7.4. The RREF of $[A \mid b]$ is

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

List pivot and free variables; write the general solution.

C7.5. Decide unique / none / infinitely many from each RREF, with one reason.

(a) $\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right]$

(b) $\left[\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 0 & 2 \end{array}\right]$

(c) $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 5 \end{array}\right]$

C7.6. True or false, with a one-line reason: solving $Ax = b$ asks whether b is in the span of the columns of A.

C7.7. You obtain $x = (1, 2)$ from software for a 2×2 system. Compute what residual you should check, and say what "success" looks like numerically.

C7.8 (cumulative, light). Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. What geometric map is $x \mapsto Ax$? (Week 7 refresh.)

Textbook Core

- B 8.6(a), 8.9, 8.10, 8.13, 8.15
- L Ch. 2 Ex. 1: Gaussian-elimination and back-substitution portions
- L Ch. 2 Ex. 3: Compare a direct solution and RREF
- L Ch. 2 Ex. 5: Inconsistent and infinite-solution cases

Stretch

B 8.4(a–b), B 8.16, L Ch. 2 Ex. 6.

What a quiz item might change

The right-hand side with the same coefficient matrix, an RREF whose original solution set you describe, or software output with a small residual.